CATEGORIES FIBRED IN GROUPOIDS REPRESENTABLE BY ALGEBRAIC SPACES ARE DELIGNE-MUMFORD STACKS

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Editorial note. No omission markers in-span (the source notes representability by schemes as an unused aside). The two same-chapter inputs are not carded ('Omitted.' and 'Proof omitted.') and are declared external.

Allowed standard prerequisites (editorial).

- Equivalence invariance. lemma-equivalent (carded L26681). Gives (1) iff (2).
- Setoid stack characterization. Stacks, lemma stack-in-setoids-characterize (cross-chapter, named). S_F is a stack in sets.
- Morphism of spaces gives representable morphism. lemma-morphism-spaces-gives-representable-by-spaces (same chapter, NOT carded, declared external). Makes the diagonal representable.
- Presheaf transfer of etale. lemma-map-presheaves-representable-by-spaces-transformation-property (same chapter, NOT carded, declared external). Transfers surjective etale h_U -> F to the associated 1-morphism.

Source reference. Original lemma 03YQ: sources/stacks-algebraic/pinned-original-source.tex, source lines 1624–1624. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045A: sources/stacks-algebraic/pinned-original-source.tex, source lines 1118–1118. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02ZZ: sources/stacks-algebraic/pinned-original-source.tex, source lines 823–823. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0432: focused-seeds/geometry/S9-alg-representable-algebraic/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}.*

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *A category fibred in groupoids $p : \mathcal{X} \to (Sch/S)_{fppf}$ which is representable by an algebraic space is a Deligne-Mumford stack.*
(2) *If F is an algebraic space over S, then the associated category fibred in groupoids $p : \mathcal{S}_F \to (Sch/S)_{fppf}$ is a Deligne-Mumford stack.*
(3) *If $X \in \mathrm{Ob}((Sch/S)_{fppf})$, then $(Sch/X)_{fppf} \to (Sch/S)_{fppf}$ is a Deligne-Mumford stack.*

Proof. It is clear that (2) implies (3). Parts (1) and (2) are equivalent by Lemma 03YQ. Hence it suffices to prove (2). First, we note that $\mathcal{S}_F$ is stack in sets since F is a sheaf (Stacks, Lemma 0432). A fortiori it is a stack in groupoids. Second the diagonal morphism $\mathcal{S}_F \to \mathcal{S}_F \times \mathcal{S}_F$ is the same as the morphism $\mathcal{S}_F \to \mathcal{S}_{F \times F}$ which comes from the diagonal of F. Hence this is representable by algebraic spaces according to Lemma 02ZZ. Actually it is even representable (by schemes), as the diagonal of an algebraic space is representable, but we do not need this. Let U be a scheme and let $h_U \to F$ be a surjective étale morphism. We may think of this as a surjective étale morphism of algebraic spaces. Hence by Lemma 045A the corresponding 1-morphism $(Sch/U)_{fppf} \to \mathcal{S}_F$ is surjective and étale. □